Venkata Sai

Sr Software Engineer

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Persistent Systems | High-Scale Monorepo Infrastructure for Developer Platforms

- Designed **Go**-based services for large-scale **monorepo** infrastructure, enhancing developer experience for 2000+ engineers, optimizing code retrieval by 35%, ensuring high performance.
- Led development of automated **Go** tooling for **code review** and **CI/CD** pipelines, using systems like **Buck** and **Gradle**, ensuring a consistently green main branch and 99.99% system stability.
- Architected a custom **SubmitQueue**-like system using **Go** and **Bazel** for managing daily diffs, improving developer productivity by 20% across the large engineering organization.
- Defined and enforced **monitoring** and **alerting** standards using **Prometheus** and **Grafana** for critical services, reducing incident response time by 40% through effective remediation.
- Owned projects end-to-end, making sound technical decisions under ambiguity, balancing engineering rigor with business priorities, including pragmatic build-vs-buy and rewrite-vs-incremental tradeoffs.
- Enhanced platform **scalability** and **reliability** through rigorous **testing** strategies and performance optimizations in **Go** microservices, reducing service latency by 25% for 5M daily users.
- Mentored junior engineers, fostering learning culture through effective communication and cross-functional collaboration, using **SonarQube** and detailed **documentation** to improve code quality 18%.

Software Engineer *Jul 2022 – Jun 2025*

Nagarro | Enterprise DevOps Tooling

- Developed and maintained core **Go** microservices using **Docker** and **Kubernetes**, ensuring high availability and processing 100,000+ requests/second with 99.9% uptime and reliability.
- Automated deployment processes using **Jenkins** and **Docker** for **CI/CD pipelines**, reducing deployment time by 83% (30 to 5 minutes), boosting developer productivity by 25%.
- Implemented comprehensive **monitoring** and **alerting** solutions with **AWS CloudWatch** for backend services, proactively identifying and resolving issues 30% faster, improving overall system health.
- Wrote extensive unit and integration **testing** for new features in **Python** and **Go**, ensuring high code quality, achieving 95% test coverage and operational readiness quickly.
- Designed and implemented high-performance **gRPC** APIs for internal service communication, handling data transfer for 50+ internal applications with 2x throughput, serving 1M transactions daily.
- Optimized database queries and schema in **PostgreSQL**, enhancing application performance by 20% and reducing average query response time by 100ms for critical operations globally.
- Deployed serverless functions using **AWS Lambda** and managed data storage on **AWS S3** for critical data processing tasks, reducing infrastructure costs by 15% annually for the platform.
- Actively participated in **code review** processes, providing constructive feedback and contributing to technical **documentation**, fostering collaboration across cross-functional teams for 10% maintainability.

Software Engineer *Jul 2020 – Aug 2022*

Virtusa | Enterprise Application Integration

- Developed backend services using **Java** and **Spring Boot** for an enterprise application, integrating 3rd-party APIs and serving 5,000+ daily active users, enhancing core functionalities significantly.
- Built and maintained **RESTful APIs** using **Python** to facilitate data exchange between internal systems, reducing manual data synchronization efforts by 25% across business departments.
- Managed and optimized relational databases using **MySQL** for application data, ensuring data integrity and improving query response times by 15% for key business reports.
- Implemented unit and integration **testing** using **JUnit** and **Mockito** for critical application modules, improving code quality and reducing bug count by 20% in production environments.
- Assisted in deploying and managing applications on **Azure** Virtual Machines and App Services, ensuring continuous availability with 99.5% uptime, supporting 10,000+ daily users effectively.

Skills

Languages: Go, Python, Java
Frameworks & Tools: Docker, Kubernetes, Jenkins, Prometheus, Grafana, gRPC, REST, JUnit, Spring Boot, Bazel, Buck, Gradle, Mockito, SonarQube
Cloud & Infra: AWS, Azure, AWS CloudWatch, AWS Lambda, AWS S3
Databases & Search: PostgreSQL, MySQL
Methodologies: Distributed Systems, Microservices, CI/CD, Agile, DevOps, System Design, Scalability, Reliability, Observability, Code Review, Monorepo, Project Ownership, Stakeholder Alignment, Operational Readiness, Incident Response, Testing, Mentorship, Documentation

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India